

50 Ω stranded micro-coax · cryostat wiring

Insertion Loss & Heat-Load Projections

Two 50 Ω candidates for a 208-line harness — **Samtec TCF-3450F** and **NEW N12-42M-100006-1** — projected across five assembly lengths: **3", 5", 10", 15", 18"**.

Samtec TCF-3450F

34 AWG (7×42) SPC / FEP-foam / 46 AWG SPC serve shield, 50 Ω

Z ₀	50.5 Ω	
Cable OD	0.56 mm	22.1 mils
α @ 2 GHz	0.637 dB/m	

NEW N12-42M-100006-1

34 AWG (7×42) C27000 brass / PFA / 44 AWG brass spiral shield, 50 Ω

Z ₀	49.8 Ω	
Cable OD	0.71 mm	28.0 mils
α @ 2 GHz	4.39 dB/m	

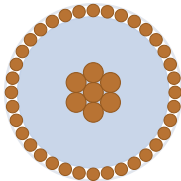
Method — conductor loss after Pozar (Microwave Engineering, §2.7) plus dielectric loss, with a per-conductor stranding factor K_s calibrated to the TCF-3450F insertion-loss curve over 1–4 GHz. Heat load is steady-state conduction across each cryostat stage's ΔT, with lay-angle path corrections. Insertion loss is quoted at the 40 K → 4 K stage ($\bar{T} \approx 22$ K); cross-sections are drawn to each preset's real geometry at a common scale.

3" assembly · 76 mm

PER-CABLE FIGURES · 50 Ω · 40 K → 4 K STAGE

Samtec TCF-3450F

34 AWG (7×42) SPC / FEP-foam / 46 AWG SPC serve shield, 50 Ω



Z_0 50.5 Ω

OD 0.56 mm · 22.1 mils

RF INSERTION LOSS

2 GHz

0.049 dB

4 GHz

0.069 dB

CONDUCTED HEAT LOAD · PER CABLE

40 K → 4 K

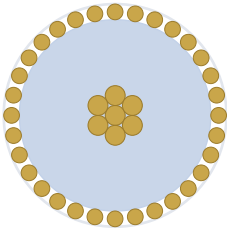
35.3 mW

4 K → 1 K

566 μW

NEW N12-42M-100006-1

34 AWG (7×42) C27000 brass / PFA / 44 AWG brass spiral shield, 50 Ω



Z_0 49.8 Ω

OD 0.71 mm · 28.0 mils

RF INSERTION LOSS

2 GHz

0.335 dB

4 GHz

0.475 dB

CONDUCTED HEAT LOAD · PER CABLE

40 K → 4 K

533 μW

4 K → 1 K

5.1 μW

Trade-off · TCF-3450F 6.9× lower loss · N12-42M 66× lower heat

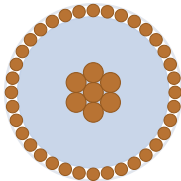
Both cables are 50 Ω. Model — Pozar conductor loss + dielectric loss with a per-conductor stranding factor K_s calibrated to the TCF-3450F insertion-loss curve (1–4 GHz); heat load is steady-state conduction across each stage's ΔT with lay-angle path corrections. Insertion loss is quoted at the 40 K → 4 K stage. The two ratios are length-independent — they are the same on every length slide.

5" assembly · 127 mm

PER-CABLE FIGURES · 50 Ω · 40 K → 4 K STAGE

Samtec TCF-3450F

34 AWG (7×42) SPC / FEP-foam / 46 AWG SPC serve shield, 50 Ω



Z₀ 50.5 Ω

OD 0.56 mm · 22.1 mils

RF INSERTION LOSS

2 GHz

0.081 dB

4 GHz

0.115 dB

CONDUCTED HEAT LOAD · PER CABLE

40 K → 4 K

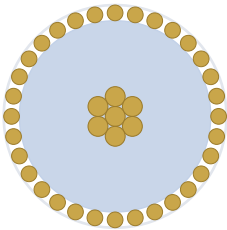
21.2 mW

4 K → 1 K

339 μW

NEW N12-42M-100006-1

34 AWG (7×42) C27000 brass / PFA / 44 AWG brass spiral shield, 50 Ω



Z₀ 49.8 Ω

OD 0.71 mm · 28.0 mils

RF INSERTION LOSS

2 GHz

0.558 dB

4 GHz

0.791 dB

CONDUCTED HEAT LOAD · PER CABLE

40 K → 4 K

320 μW

4 K → 1 K

3.1 μW

Trade-off · TCF-3450F 6.9× lower loss · N12-42M 66× lower heat

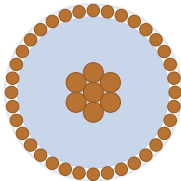
Both cables are 50 Ω. Model — Pozar conductor loss + dielectric loss with a per-conductor stranding factor K_s calibrated to the TCF-3450F insertion-loss curve (1–4 GHz); heat load is steady-state conduction across each stage's ΔT with lay-angle path corrections. Insertion loss is quoted at the 40 K → 4 K stage. The two ratios are length-independent — they are the same on every length slide.

10" assembly · 254 mm

PER-CABLE FIGURES · 50 Ω · 40 K → 4 K STAGE

Samtec TCF-3450F

34 AWG (7×42) SPC / FEP-foam / 46 AWG SPC serve shield, 50 Ω



Z_0 50.5 Ω

OD 0.56 mm · 22.1 mils

RF INSERTION LOSS

2 GHz

0.162 dB

4 GHz

0.230 dB

CONDUCTED HEAT LOAD · PER CABLE

40 K → 4 K

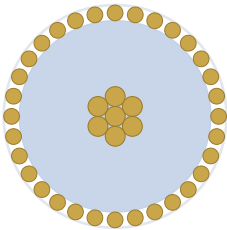
10.6 mW

4 K → 1 K

170 μW

NEW N12-42M-100006-1

34 AWG (7×42) C27000 brass / PFA / 44 AWG brass spiral shield, 50 Ω



Z_0 49.8 Ω

OD 0.71 mm · 28.0 mils

RF INSERTION LOSS

2 GHz

1.12 dB

4 GHz

1.58 dB

CONDUCTED HEAT LOAD · PER CABLE

40 K → 4 K

160 μW

4 K → 1 K

1.5 μW

Trade-off · TCF-3450F 6.9× lower loss · N12-42M 66× lower heat

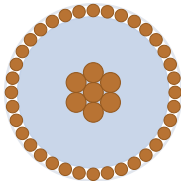
Both cables are 50 Ω. Model — Pozar conductor loss + dielectric loss with a per-conductor stranding factor K_s calibrated to the TCF-3450F insertion-loss curve (1–4 GHz); heat load is steady-state conduction across each stage's ΔT with lay-angle path corrections. Insertion loss is quoted at the 40 K → 4 K stage. The two ratios are length-independent — they are the same on every length slide.

15" assembly · 381 mm

PER-CABLE FIGURES · 50 Ω · 40 K → 4 K STAGE

Samtec TCF-3450F

34 AWG (7×42) SPC / FEP-foam / 46 AWG SPC serve shield, 50 Ω



Z₀ 50.5 Ω

OD 0.56 mm · 22.1 mils

RF INSERTION LOSS

2 GHz

0.243 dB

4 GHz

0.346 dB

CONDUCTED HEAT LOAD · PER CABLE

40 K → 4 K

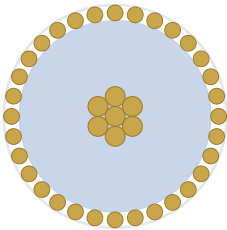
7.07 mW

4 K → 1 K

113 μW

NEW N12-42M-100006-1

34 AWG (7×42) C27000 brass / PFA / 44 AWG brass spiral shield, 50 Ω



Z₀ 49.8 Ω

OD 0.71 mm · 28.0 mils

RF INSERTION LOSS

2 GHz

1.67 dB

4 GHz

2.37 dB

CONDUCTED HEAT LOAD · PER CABLE

40 K → 4 K

107 μW

4 K → 1 K

1.0 μW

Trade-off · TCF-3450F 6.9× lower loss · N12-42M 66× lower heat

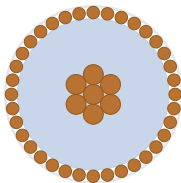
Both cables are 50 Ω. Model — Pozar conductor loss + dielectric loss with a per-conductor stranding factor K_s calibrated to the TCF-3450F insertion-loss curve (1–4 GHz); heat load is steady-state conduction across each stage's ΔT with lay-angle path corrections. Insertion loss is quoted at the 40 K → 4 K stage. The two ratios are length-independent — they are the same on every length slide.

18" assembly · 457 mm

PER-CABLE FIGURES · 50 Ω · 40 K → 4 K STAGE

Samtec TCF-3450F

34 AWG (7×42) SPC / FEP-foam / 46 AWG SPC serve shield, 50 Ω



Z_0 50.5 Ω

OD 0.56 mm · 22.1 mils

RF INSERTION LOSS

2 GHz

0.291 dB

4 GHz

0.415 dB

CONDUCTED HEAT LOAD · PER CABLE

40 K → 4 K

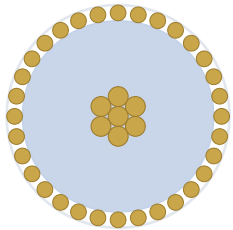
5.89 mW

4 K → 1 K

94 μW

NEW N12-42M-100006-1

34 AWG (7×42) C27000 brass / PFA / 44 AWG brass spiral shield, 50 Ω



Z_0 49.8 Ω

OD 0.71 mm · 28.0 mils

RF INSERTION LOSS

2 GHz

2.01 dB

4 GHz

2.85 dB

CONDUCTED HEAT LOAD · PER CABLE

40 K → 4 K

89 μW

4 K → 1 K

0.8 μW

Trade-off · TCF-3450F 6.9× lower loss · N12-42M 66× lower heat

Both cables are 50 Ω. Model — Pozar conductor loss + dielectric loss with a per-conductor stranding factor K_s calibrated to the TCF-3450F insertion-loss curve (1–4 GHz); heat load is steady-state conduction across each stage's ΔT with lay-angle path corrections. Insertion loss is quoted at the 40 K → 4 K stage. The two ratios are length-independent — they are the same on every length slide.

All five lengths at a glance

FULL HARNESS · 208 (13 cables of 16 lines)

Length	Samtec TCF-3450F					NEW N12-42M-100006-1				
	LOSS	Q̇ PER CABLE		208 (13 CABLES OF 16 LINES)		LOSS	Q̇ PER CABLE		208 (13 CABLES OF 16 LINES)	
	2 GHz	40→4 K	4→1 K	40→4 K	4→1 K	2 GHz	40→4 K	4→1 K	40→4 K	4→1 K
3"	0.049 dB	35.3 mW	566 μW	7.35 W	117.6 mW	0.335 dB	533 μW	5.1 μW	110.9 mW	1.06 mW
5"	0.081 dB	21.2 mW	339 μW	4.41 W	70.6 mW	0.558 dB	320 μW	3.1 μW	66.6 mW	636 μW
10"	0.162 dB	10.6 mW	170 μW	2.20 W	35.3 mW	1.12 dB	160 μW	1.5 μW	33.3 mW	318 μW
15"	0.243 dB	7.07 mW	113 μW	1.47 W	23.5 mW	1.67 dB	107 μW	1.0 μW	22.2 mW	212 μW
18"	0.291 dB	5.89 mW	94 μW	1.22 W	19.6 mW	2.01 dB	89 μW	0.8 μW	18.5 mW	177 μW

Insertion loss quoted at 2 GHz, 4–40 K stage — see the per-length slides for the 4 GHz figures. Heat load is the conducted $\dot{Q}$ across each stage's ΔT , shown per cable and for the full harness (208 (13 cables of 16 lines)). The bundle columns are the per-cable figure × 208; they are the ones that set the cryostat budget.